IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

REINFORCEMENT LEARNING FOR COMMUNICATION SYSTEMS

Friday 5 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : Prof Kin Leung
Second Marker(s) :

© Imperial College London

1. **Dynamic Programming** (25%)

A Markov Decision Process (MDP) with three states {A, B, C} is presented in Figure 1. The actions, state transitions and rewards are also shown in the figure. For example, when action i is taken in state A, the reward is +6 and the system state transits to C or back to A with probability ¾ and ¼, respectively. If action j is taken while in state A, the reward is -4 and the state is changed to B with probability 1. The discount factor is γ. The return for any given state s is the total of discounted rewards over time starting from state s by following an action policy π.

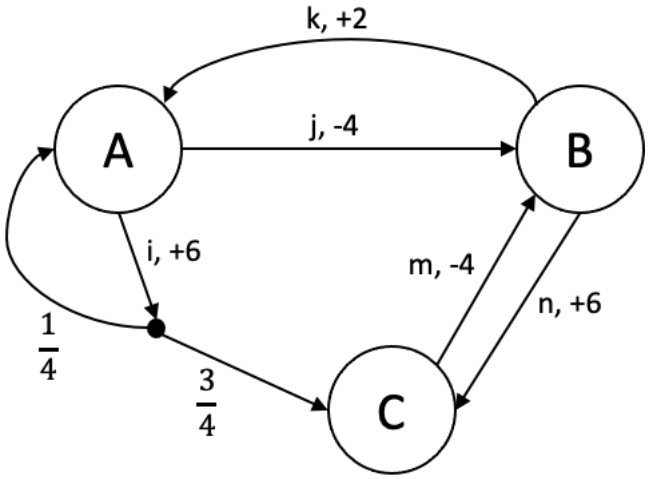


Figure 1. A simple Markov Decision Process (MDP)

(a) Construct a table to represent the above MDP. Let s, a, s', r and $p(s'|s,a)$ denote the current state, action, next state, reward and transition probability from s to s' given current state s and action a taken. The table should have columns for s, a, s', r and $p(s'|s,a)$, and one row for every 4-tupe of (s,a,s',r) for which $p(s'|s,a) > 0$. [5]

(b) Give the Bellman optimality equation for $q_*(s,a)$ when state $s = A$ and action $a = i$. To expand one time step is sufficient. [4]

(c) Consider the uniform random policy $\pi(s,a)$ that takes all possible actions from the current state with equal probability. For example, $\pi(A,i) = \pi(A,j) = 0.5$. Initialize the state value to be 0 for all states. Perform one synchronous round of policy evaluation iteration. [5]

(d) Using the updated state-value function from part (c), apply one iteration of greedy policy improvement to obtain an "optimal" deterministic policy $\pi_g(s)$ for each state $s = A, B$ or C. Use $\gamma = 0.9$. [5]

(e) Initialize the state value to be 0 for all states. Perform two synchronous rounds of value iteration and fill in the following table with the appropriate values. Use $\gamma = 0.9$. [6]

State	$q(s,a)$					Best action	$v(s) = \max_a q(s,a)$
	i	j	k	m	n		
A							
B							
C							
A							
B							
C							

2. Batch TD(0) Learning (25%)

Consider an undiscounted ($\gamma = 1$) Markov Reward Process (MRP) with two states A and B. The transition matrix and reward function are unknown, but three sample episodes have been observed:

(i) A+4 → A+3 → B−4 → A+3 → B−3 → terminate

(ii) B−1 → A+3 → B−2 → terminate

(iii) B−2 → A+3 → A+2 → terminate

In the above episodes, state transitions and rewards are shown at each step. For example, A+3 → A in the last episode (iii) indicates a transition from state A to state A with a reward of +3.

(a) Use the first-visit Monte-Carlo (MC) method to estimate the state-value functions of V(A) and V(B). [5]

(b) Use the every-visit MC method to estimate the state-value functions of V(A) and V(B). [5]

(c) Assume that the three episodes are repeatedly used in the learning process. As a result, the episodes sufficiently capture the transition probabilities between states A, B and the terminal state. Obtain these state transition probabilities. [5]

(d) What are the average rewards associated with state A and B, as reflected in the three episodes? [5]

(e) Using the state transition probabilities and the average state rewards in parts (c) and (d), give the Bellman expectation equations for the MRP. Solve the values of V(A) and V(B) for the MRP. [5]

3. Q-Learning and Sarsa (20%)

Consider a system with four states {A, B, C, D}, two possible actions {i, j} and rewards as Figure 2 shows. For example, when action i is taken in state A, the reward is -4, and the state changes from A to B with probability 1.

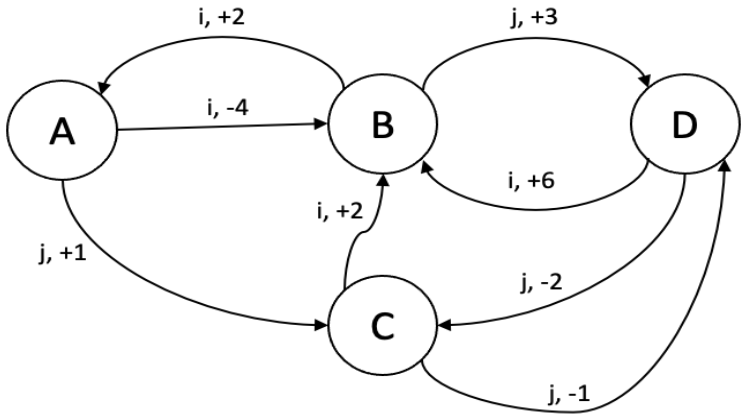


Figure 2: A four-state system.

(a) Explain the main difference between Q-learning algorithm and Sarsa algorithm. [4]

(b) Use the Q-learning algorithm to update Q values with the discount factor of $\gamma = 0.9$ and the learning rate of $\alpha = 0.5$. The Q values for all states and actions are initialized to 0. Consider that the system begins in state A and takes a sequence of four actions {i, j, j, i}, and the associated action-value functions are updated according to the Q-learning algorithm. Fill the following table of Q values after these updates. [8]

Q(s,a)	A	B	C	D
i				
j				

(c) Assume that Sarsa algorithm is used to update Q values for the same setting and the system also takes the sequence of actions {i, j, j, i} from state A, as in part (b). Fill the following table of Q values after these updates. Assume that action i is taken after the sequence of 4 actions and this final transition is not used to update Q values. [8]

Q(s,a)	A	B	C	D
i				
j				

4. **RL Controller for Base-Station Transmissions** (30%)

Consider two neighboring cells, *A* and *B*, with a base-station (BS) transmitter in each cell in Figure 3. By using frequency bandwidth of W Hz, the two BS transmitters simultaneously transmit data at the same power level P to two users, one associated with each of the cells. Let G_A and G_B be the large set of users associated with the respective cells.

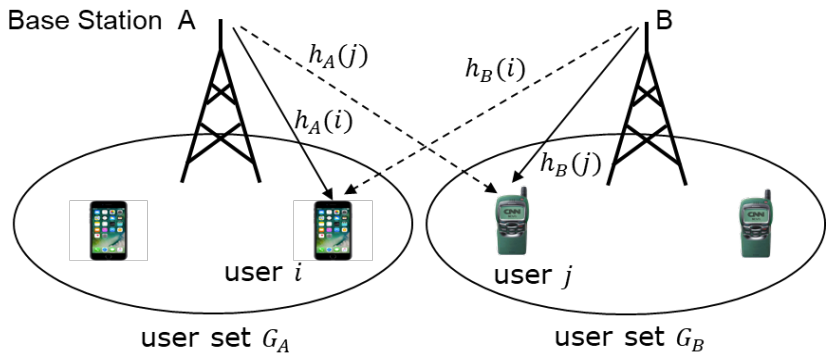


Figure 3. Downlink transmissions in two neighboring cells A and B.

Let system time be divided into *time slots* and 8 consecutive time slots form a *time frame* in Figure 4. Both BS transmitters simultaneously send data in each time slot to two users indexed i and j where $i \in G_A$ and $j \in G_B$.

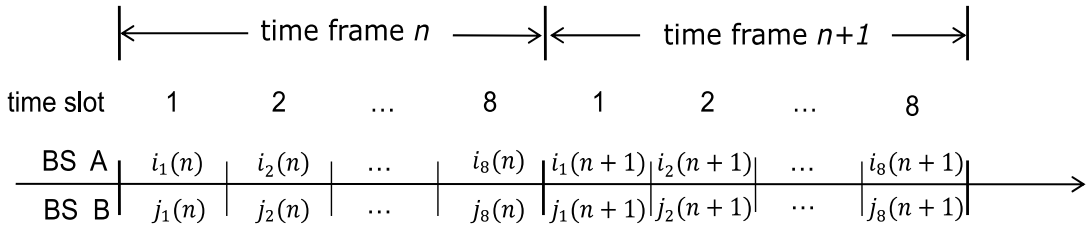


Figure 4. Transmission schedule in the time slots and time frame.

Due to radio interference (with noise neglected), the signal-to-interference ratio (SIR) α_i at user $i \in G_A$ or G_B is given by

$$\alpha_i = \frac{h_A(i)\,P}{h_B(i)\,P} \quad \text{or} \quad \alpha_i = \frac{h_B(i)\,P}{h_A(i)P}\ ,$$

respectively, where P is the transmission power, and $h_A(i)$ and $h_B(i)$, which are greater than 0, denote the corresponding channel gain from BS transmitter A and B to user i. Assume that $h_A(i)$ and $h_B(i)$ for all user i can be accurately estimated in advance and do not change when the users are receiving data from the BS's over a number of time frames. By the Shannon capacity formula, data rate R_i for user $i \in G_A$ or G_B is

$$R_i = \mathrm{W}\log_2\alpha_i = \mathrm{W}\log_2\left(\frac{h_A(i)}{h_B(i)}\right) \quad \text{or} \quad R_i = W\log_2\alpha_i = \mathrm{W}\log_2\left(\frac{h_B(i)}{h_A(i)}\right),$$

respectively. With probability σ, the concurrent transmissions from both BS transmitters in each time slot will continue in the same time slot in the next time frame. With probability $1-\sigma$, the transmissions in a time slot will complete and the same time slot in the next time frame onward will become available for transmission to two new users, one from G_A and another from G_B.

Consider that there are k time slots available for $k = 0, 1, \ldots, 8$ at the end of a time frame. We aim to design an RL controller (agent) to perform the pairing of k new users from G_A with another k

new users from G_B so that one user from G_A is paired with another user from G_B to receive data in each of the k newly available time slots starting in the next time frame. As the respective channel gains, $h_A(i)$ and $h_B(i)$, from the BS transmitters to each user i are known in advance to the RL controller, the objective of the user pairing is to minimize the radio interference due to simultaneous transmissions and thus maximize the total data rate of all continuing and new data transmissions in all time slots of each time frame, which is discounted over time frames with a discount factor γ.

Figure 4 shows the users in both cells scheduled to receive data transmitted from the BS transmitters in different time slots and time frames. For example, BS transmitter A and B send data to users $i_2(n) \in G_A$ and $j_2(n) \in G_B$, respectively, in time slot 2 during time frame n in the figure. If the BS transmissions in time slot m for $m = 1$ to 8 continue in time frame $n + 1$, $i_m(n + 1) = i_m(n)$ and $j_m(n + 1) = j_m(n)$. Use the notation in Figure 4 to answer the following questions.

(a) Define the system state S_n at the end of each time frame n for the RL controller. [7]

(b) Explain why the system with your state definition satisfies the Markovian property. [4]

(c) Define the actions of the RL controller at the end of each time frame. [5]

(d) Define the reward for each time frame n, including the new transmissions in some time slots, to achieve the objective stated above. [5]

(e) Suggest one RL algorithm to perform the pairing of users in real time for simultaneous reception of data from the BS transmitters in each available time slot and explain why it is appropriate. [4]

(f) To highlight the performance of your RL approach to pairing up users for receiving data from the BS transmitters, suggest a simple, baseline scheme for such comparison. Explain why your suggested scheme is appropriate. [5]